

SW1:

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#hostname Layer2Switch-1
```

```
Layer2Switch-1(config)#spanning-tree vlan
```

```
*Aug 13 15:36:39.279: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vios_l2-adventerprisek9-m'
passed code signing verification
```

```
Layer2Switch-1(config)#spanning-tree vlan 1 priority 4096
```

```
Layer2Switch-1(config)#exit
```

```
Layer2Switch-1#w
```

```
*Aug 13 15:36:57.810: %SYS-5-CONFIG_I: Configured from console by console
```

```
Layer2Switch-1#write memory
```

Building configuration...

Compressed configuration from 5281 bytes to 2030 bytes[OK]

```
Layer2Switch-1#
```

```
*Aug 13 15:37:08.195: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk.
Please wait...
```

```
*Aug 13 15:37:08.991: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk
successfully.
```

```
Layer2Switch-1#
```

SW2:

```
vIOS-L2-01>enable
```

```
vIOS-L2-01#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
vIOS-L2-01(config)#hostname Layer2Switch-2
```

```
Layer2Switch-2(config)#spanning-tree vlan 1 priority 8192
```

Building configuration...

*Aug 13 15:41:12.327: %SYS-5-CONFIG_I: Configured from console by consoleCompressed configuration from 5281 bytes to 2031 bytes[OK]

Layer2Switch-2(config)#exit

Layer2Switch-2#write memory

*Aug 13 15:41:18.671: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Aug 13 15:41:19.494: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

SW3:

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#hostname Layer2Switch-3

Layer2Switch-3(config)#spanning-tree vlan 1

Layer2Switch-3(config)#exit

*Aug 13 15:44:23.493: %SYS-5-CONFIG_I: Configured from console by console

Layer2Switch-3#write memory

Building configuration...

Compressed configuration from 5046 bytes to 1924 bytes[OK]

*Aug 13 15:44:35.353: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...

*Aug 13 15:44:36.150: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.

Layer2Switch-3#

SW4:

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#hostname Layer2Switch-4

Layer2Switch-4(config)#spanning-tree vlan 1

Layer2Switch-4(config)#exit

Layer2Switch-4#w

*Aug 13 15:47:10.667: %SYS-5-CONFIG_I: Configured from console by console

Layer2Switch-4#write memory

Building configuration...

Compressed configuration from 5046 bytes to 1919 bytes[OK]

Layer2Switch-4#

*Aug 13 15:47:21.577: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk.
Please wait...

*Aug 13 15:47:22.373: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk
successfully.

SW5:

vIOS-L2-01>enable

vIOS-L2-01#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

vIOS-L2-01(config)#hostname Layer2Switch-5

Layer2Switch-5(config)#spanning-tree vlan 1

Layer2Switch-5(config)#exit

Layer2Switch-5#

*Aug 13 15:48:18.095: %SYS-5-CONFIG_I: Configured from console by console

Layer2Switch-5#write memory

Building configuration...

Compressed configuration from 5046 bytes to 1926 bytes[OK]

*Aug 13 15:48:30.164: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk.
Please wait...

*Aug 13 15:48:30.959: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk
successfully.

Layer2Switch-5#

ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=6.753 ms

84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=7.019 ms

84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.996 ms

84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=6.278 ms

84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=7.397 ms

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=4.425 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=8.108 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=6.601 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=8.170 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=7.542 ms

PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=15.606 ms

84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=10.272 ms

84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=5.053 ms

84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=9.669 ms

84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=7.836 ms

PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=18.151 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=5.405 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=7.781 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=6.355 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=16.871 ms

PC1> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=9.552 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=6.480 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=14.855 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=6.448 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=7.096 ms

PC1>

ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=9.430 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=2.176 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=4.399 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=7.402 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=14.841 ms

PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=11.370 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=5.249 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=4.431 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=3.227 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=7.560 ms

PC2> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=5.784 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=10.774 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=8.273 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=6.618 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=4.798 ms

PC2> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=16.423 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=15.350 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=13.730 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=2.806 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=6.158 ms

PC2>

PC3> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=1.272 ms

84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=0.951 ms

84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=1.010 ms

84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=4.822 ms

84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=6.947 ms

PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=15.606 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=7.619 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=7.841 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=6.489 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=9.715 ms

PC3> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=21.061 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=9.140 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=4.044 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=5.324 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=6.706 ms

PC3>

ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=7.894 ms

84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=12.585 ms

84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=5.006 ms

84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=8.354 ms

84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=12.321 ms

PC4> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=14.428 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=4.895 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=3.512 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=10.309 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=4.707 ms

PC4>

PC5> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=1.081 ms

84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=6.963 ms

84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=7.296 ms

84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=1.908 ms

84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=7.645 ms

PC5>

SW1 → SW2

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f9:28:9b:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f9:28:9b:00:00

Port identifier: 0x8001

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

SW1 → SW3

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f9:28:9b:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f9:28:9b:00:00

Port identifier: 0x8004

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

SW1 → SW4

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f9:28:9b:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f9:28:9b:00:00

Port identifier: 0x8005

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

SW1 → SW5

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:f9:28:9b:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:f9:28:9b:00:00

Port identifier: 0x8007

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

Layer2Switch-3#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Layer2Switch-3(config)#interface GigabitEthernet0/1

Layer2Switch-3(config-if)#spanning-tree cost 1000

Layer2Switch-3(config-if)#exit

Layer2Switch-3(config)#exit

*Aug 13 16:31:33.814: %SYS-5-CONFIG_I: Configured from console by console

Building configuration...

Compressed configuration from 5071 bytes to 1939 bytes[OK]

Layer2Switch-3#

*Aug 13 16:31:50.288: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk.
Please wait...

*Aug 13 16:31:51.094: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk
successfully.

Layer2Switch-3#

